Determining which side

Task A: Identifying the hypotenuse

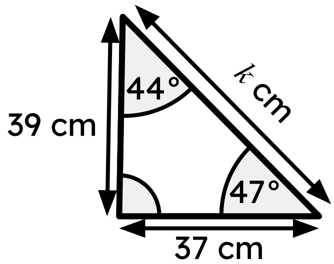
1) Complete each statement.

a) Pythagoras' theorem can only be used to calculate the length of a missing side of a triangle if one of its sides is a *hypotenuse*.

b) Only *right-angled* triangles have a hypotenuse.

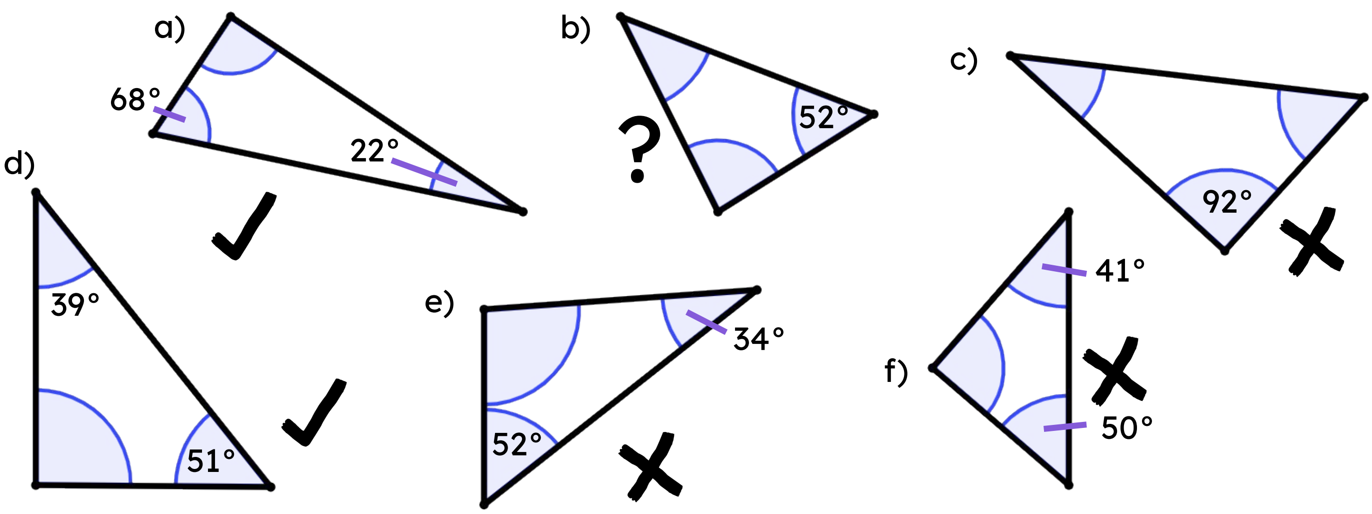
c) The hypotenuse is always opposite the *right angle*.

2) Write a sentence to explain why Pythagoras' theorem cannot be used to find the value k.



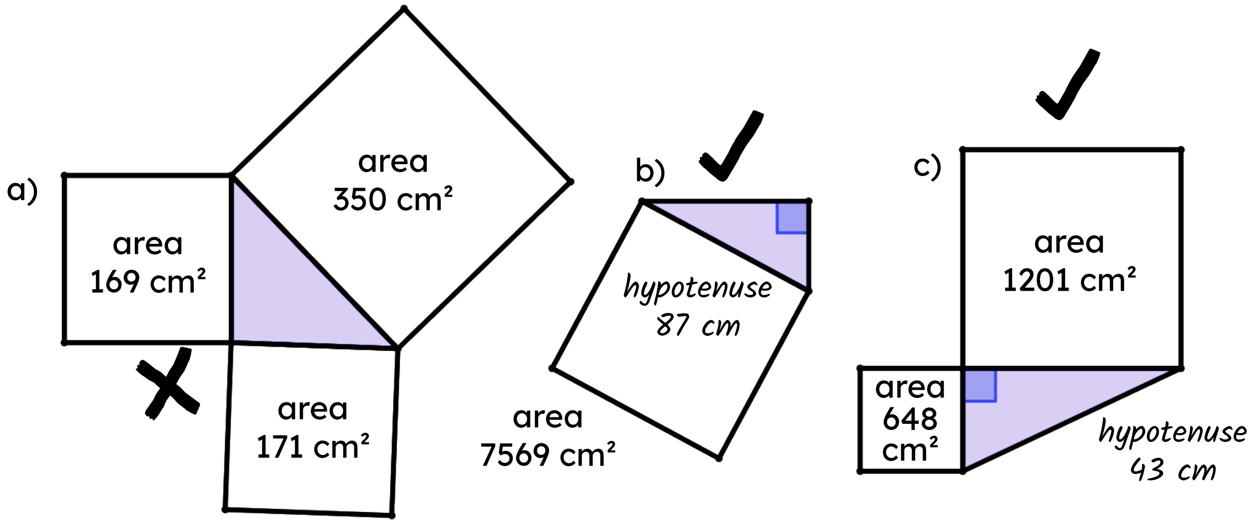
The missing angle is 89°, and therefore this triangle is not right-angled.
You cannot use Pythagoras' theorem on triangles that are not right-angled.

3) For each of these triangles, tick (✔) each right-angled triangle, cross (✘) non-right-angled triangle, put a question mark (?) by each one where you cannot tell. For each right-angled triangle, circle or highlight its hypotenuse.



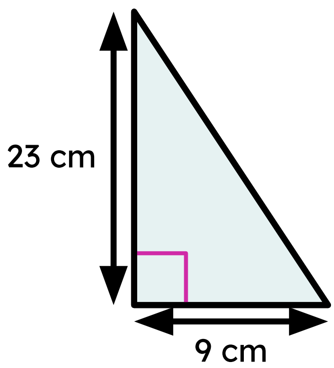
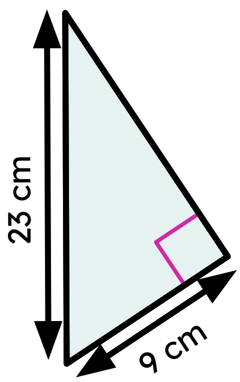
4) For which of these triangles can you use Pythagoras' theorem?

Where possible, calculate the length of the hypotenuse and label its length.



Task B: Right-angled triangles and congruence

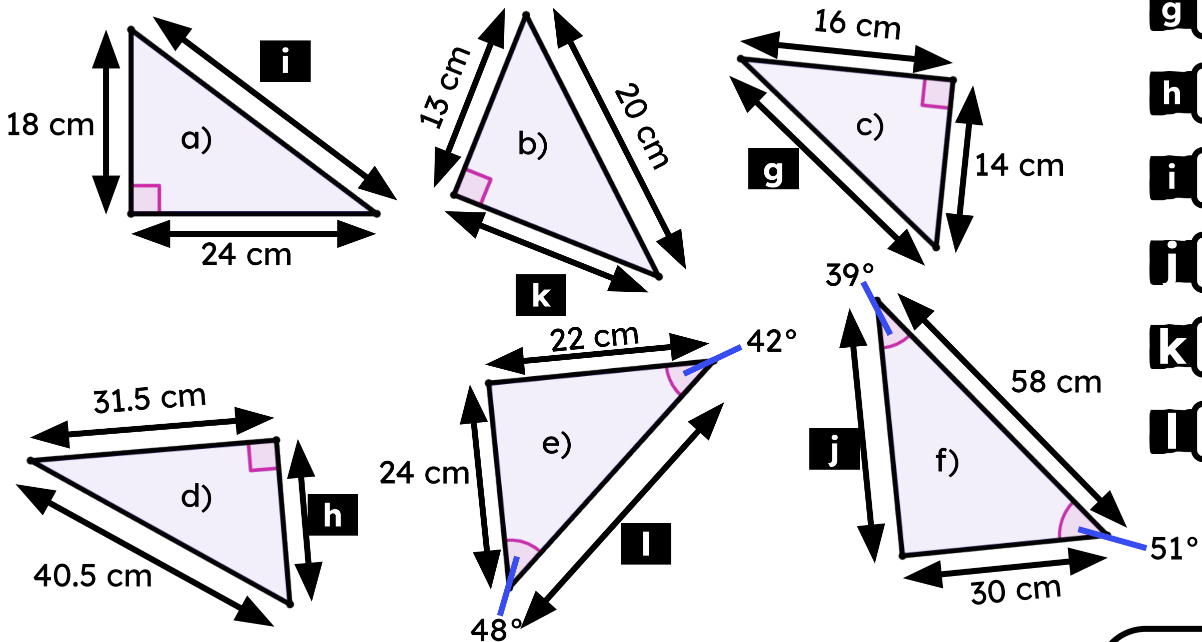
1) x represents the length of the missing side on one of these two triangles.
Match these calculations created from the Pythagoras' theorem formula to the correct triangle.

triangle A	triangle B	neither triangle
		**b** **g**
a **d**	**c** **e** **f**	

- **a** $23^2 + 9^2 = x^2$
- **b** $23^2 + x^2 = 9^2$
- **c** $x^2 + 9^2 = 23^2$
- **d** $9^2 + 23^2 = x^2$
- **e** $9^2 + x^2 = 23^2$
- **f** $x^2 = 23^2 - 9^2$
- **g** $x^2 = 9^2 - 23^2$

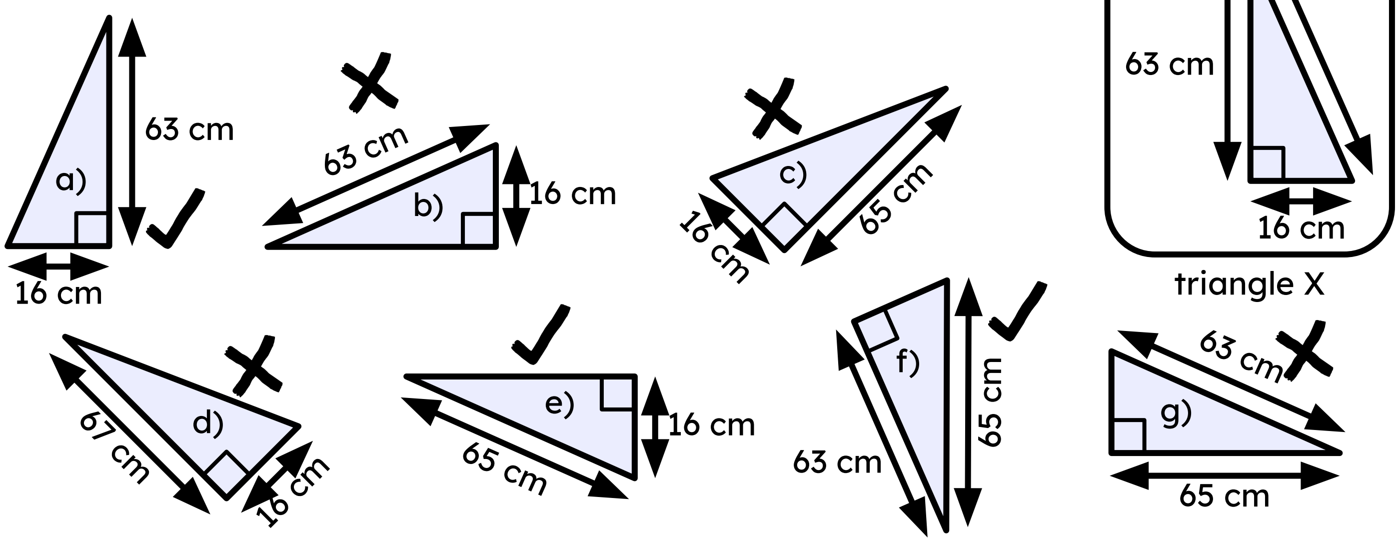
Task C: Finding the length of missing sides

1) Match the triangle to the length of its missing side.



- g: 21.26 cm
- h: 25.46 cm
- i: 30 cm
- j: 49.64 cm
- k: 15.20 cm
- l: 32.56 cm

2) Which of these triangles are congruent to triangle X?



3) Calculate the length of each unknown side labelled with a letter, rounding your answer to 2 decimal places.